

**CSES Problem Set****Tree Diameter**TASK | [SUBMIT](#) | [RESULTS](#) | [STATISTICS](#) | [TESTS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes.

The *diameter* of a tree is the maximum distance between two nodes. Your task is to determine the diameter of the tree.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print one integer: the diameter of the tree.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

3

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Your submissions

Explanation: The diameter corresponds to the path $2 \rightarrow 1 \rightarrow 3 \rightarrow 5$.